

Discrete Mathematics Assignment 5.

- Q.1) Determine an optimal tree and by using Huffman algorithm for file containing set of characters with weights also find the Huffman code for the optimal tree?

	a	t	i	m	s	x	d
frequency	3	6	5	4	16	8	10

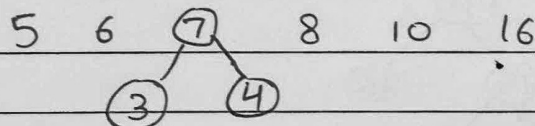
solⁿ:- According to Huffman algorithm

Step:1) Ascending order

3 4 5 6 8 10 16

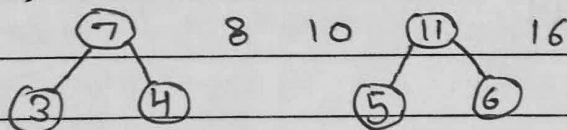
Step:2)

$$3+4=7$$



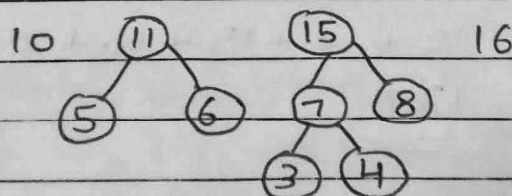
Step:3)

$$5+6=11$$



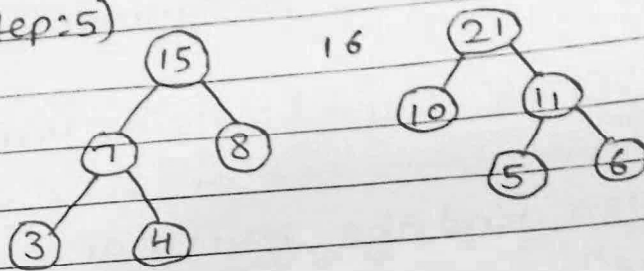
Step:4)

$$7+8=15$$



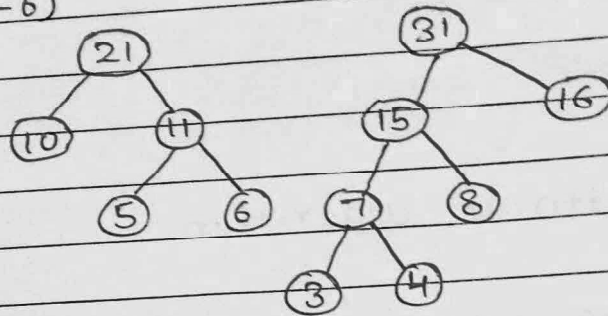
$$10 + 11 = 21$$

Step:-5)



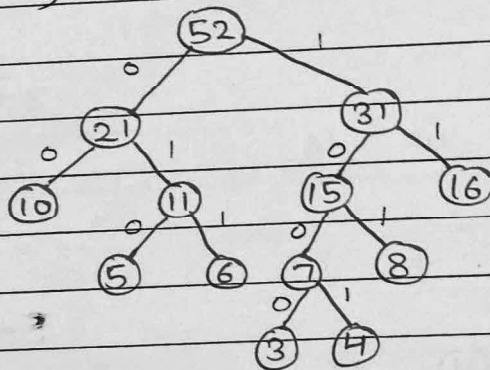
$$15 + 16 = 31$$

Step:-6)



$$21 + 31 = 52$$

Step:-7)



Prefix code

3 → 1000

4 → 1001

5 → 010

6 → 011

8 → 101

10 → 00

16 → 11

Weight of optimal tree:-

$$\Rightarrow 3 \times 4 + 4 \times 4 + 5 \times 3 + 6 \times 3 + 8 \times 3 + 10 \times 2 + 16 \times 2$$

$$\Rightarrow 12 + 16 + 15 + 18 + 24 + 20 + 32$$

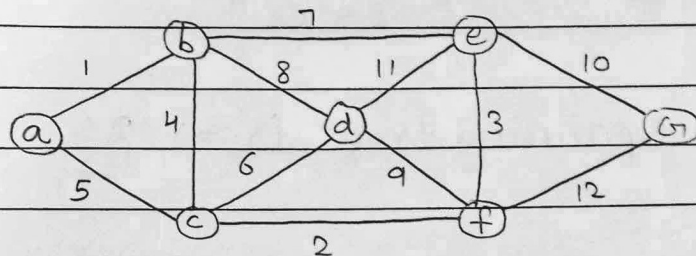
$$\Rightarrow 137$$

∴ weight of optimal tree is

$$\Rightarrow 137$$

NICE

Q.2) Use the Prim's Algorithm and find the minimum Spanning tree for the graph.

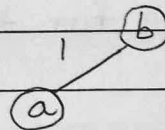


Soln:- Using Prim's algorithm:

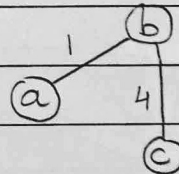
Step: 1) Select vertex a
 $MST = \{ (a) () \}$

(a)

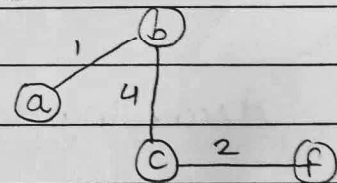
Step: 2) a connected to b
 $MST = \{ (a,b) (e) \}$



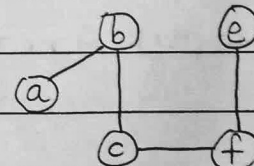
Step: 3) b connected to c
 $MST = \{ (a,b,c) (e_1, e_2) \}$



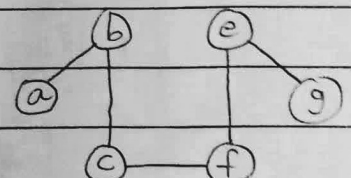
Step: 4) c connected to f
 $MST = \{ (a,b,c,f) (e_1, e_2, e_3) \}$



Step: 5) f connected e
 $MST = \{ (a,b,c,f,e) (e_1, e_2, e_3, e_4) \}$

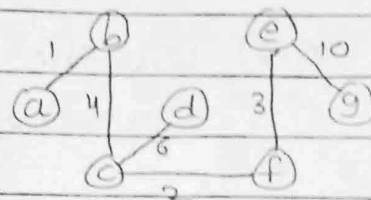


Step: 6) e connected g
 $MST = \{ (a,b,c,f,e,g) (e_1, e_2, e_3, e_4, e_5) \}$

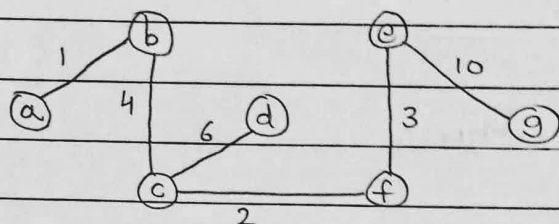


Step: 7 c connected to d

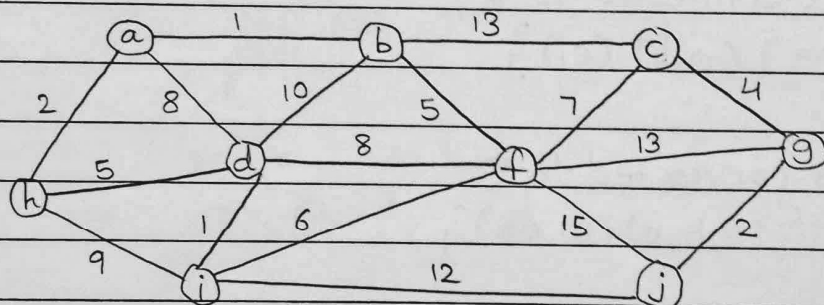
$MST = \{ (a,b,c,f,e,g,d) (e_1, e_2, e_3, e_4, e_5, e_6) \}$



$\therefore$ Minimum Spanning tree is $\Rightarrow$ 26.



Q.3) Use the Kruskal's Algorithm and find the minimum spanning tree for the graph.



Solⁿ:-

According to Kruskal's algorithm;

Step: 1) Arrange the edges in ascending order according to their weight.

ab	di	ah	gj	cg	dh	bf	fi	cf	ad	df	hi	bd	ij	bc	fg	fj
1	1	2	2	4	5	5	6	7	8	8	9	10	12	13	13	15

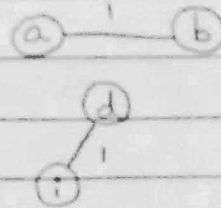
Step: 2) select a-b

$$MST = 1$$



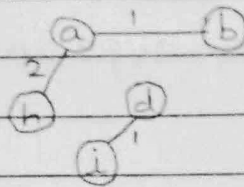
Step: 3) select d-i

$$MST = 1 + 1 = 2$$



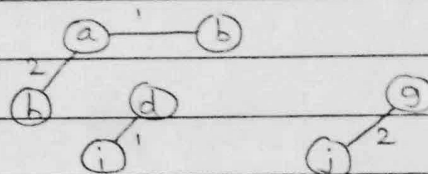
Step: 4) select a-h

$$MST = 2 + 2 = 4$$



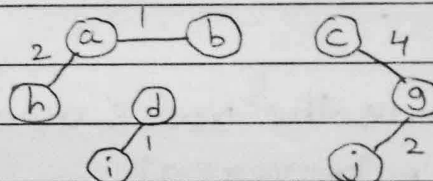
Step: 5) select g-j

$$MST = 4 + 2 = 6$$



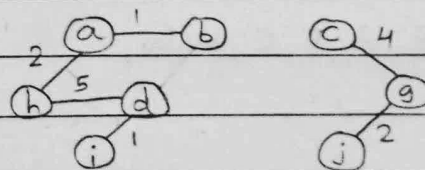
Step: 6) select c-g

$$MST = 6 + 4 = 10$$



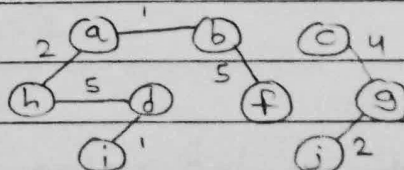
Step: 7) select d-h

$$MST = 10 + 5 = 15$$



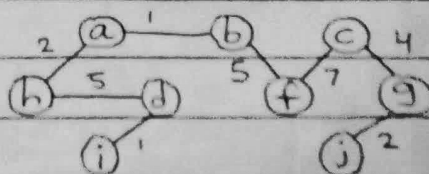
Step: 8) select b-f

$$MST = 15 + 5 = 20$$

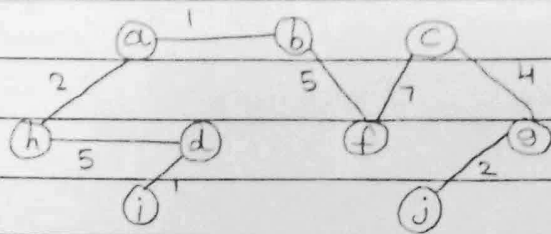


Step: 9) select c-f

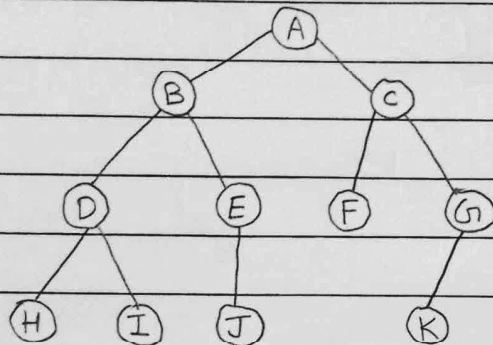
$$MST = 20 + 7 = 27$$



Minimum Spanning tree is 27.



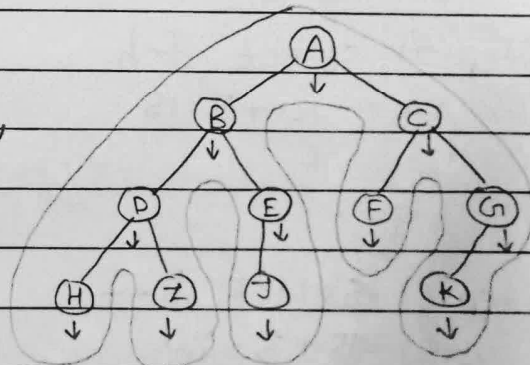
Q.4) Answer the given question Based on the given tree.



(i) Determine the Nodes using In-order Traversal.

Solⁿ:- In-order Traversal.

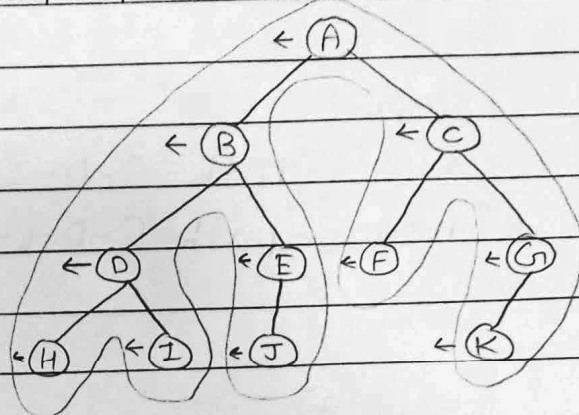
⇒ H, D, I, B, J, E, A, F, C, K, G,



(ii) Determine the nodes using pre-order Traversal.

Solⁿ:- Pre-order Traversal.

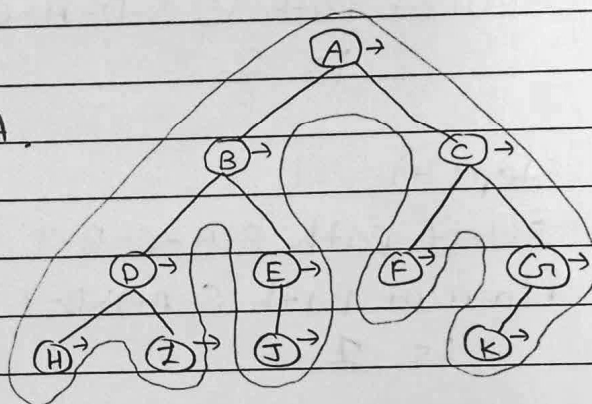
⇒ A, B, D, H, I, E, J, C, F, G, K.



(iii) Determine the nodes using post-order Traversal.

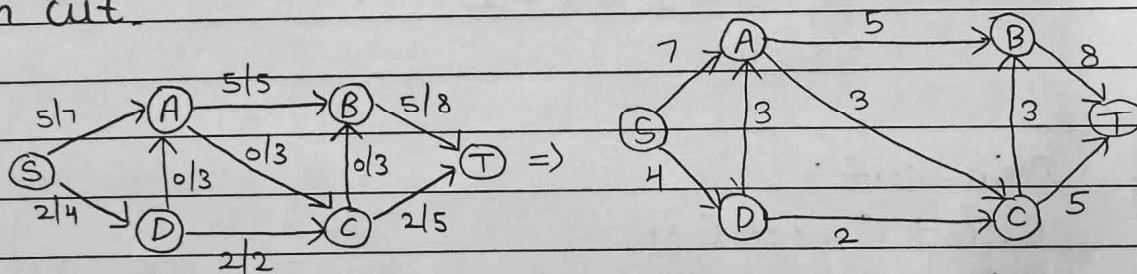
Solⁿ:- Post-order Traversal.

⇒ H, I, D, J, E, B, F, K, G, C, A.



Q.5) Find the max flow in the Given Network using Labeling Procedure and determine the corresponding Min cut.

Solⁿ:-

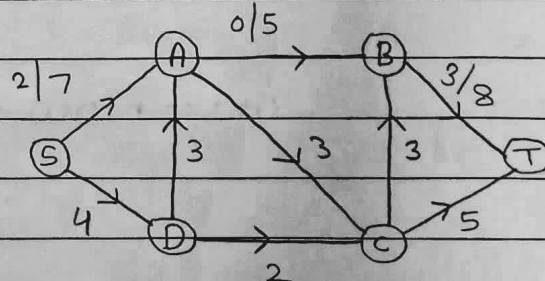


according to max flow-min cut theorem.

Step: i)

Select path S-A-B-T

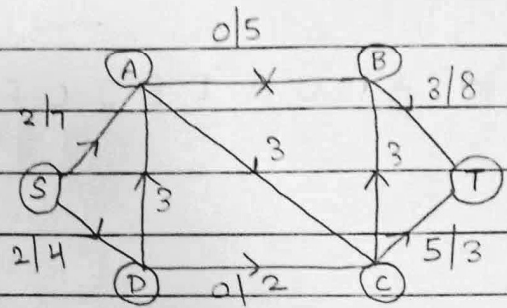
flow of path S-A-B-T is 5



Step:2)

Select path S-D-C-T

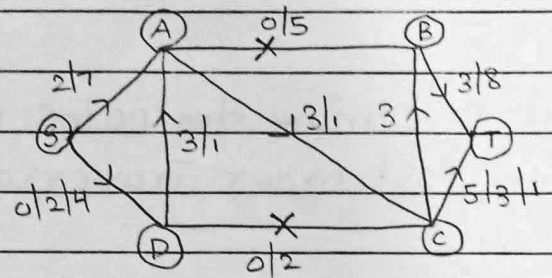
flow of path S-D-C-T is 2



Step:3)

Select path S-D-A-C-T

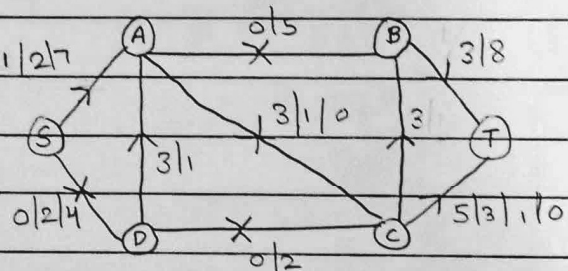
flow of path is S-D-A-C-T
is 2



Step:4)

Select path S-A-C-B-T

flow of path S-A-C-B-T
is 1



maximum Flow:-

flow of Path, S-A-B-T + S-D-C-T + S-D-A-C-T + S-A-C-B-T

$$\Rightarrow 5 + 2 + 2 + 1$$

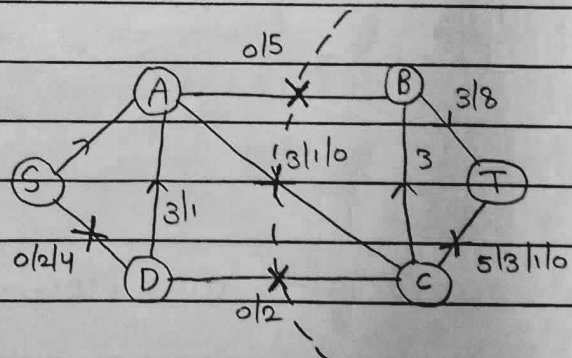
$$\Rightarrow 10$$

Min-cut:-

$C_{AB} + C_{AC} + C_{CD}$

$$5 + 3 + 2$$

$$10$$



$\therefore \text{max-flow} = \text{min-cut} = 10$